

assignment14 - Source of magnetism, Faraday's law I

⚠ This is a preview of the published version of the quiz

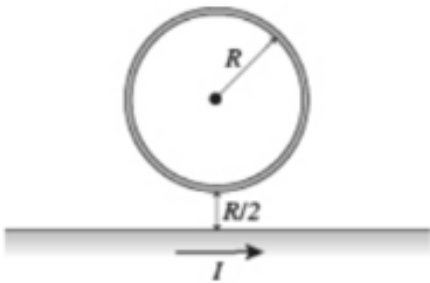
Started: Mar 23 at 6:40pm

Quiz Instructions



Question 1 15 pts

69) A single current-carrying circular loop of radius R is placed next to a long, straight wire, as shown in the figure. The current I in the wire flows to the right. In which direction must current flow in the loop to produce a net magnetic field of zero at its center?



CCW



0



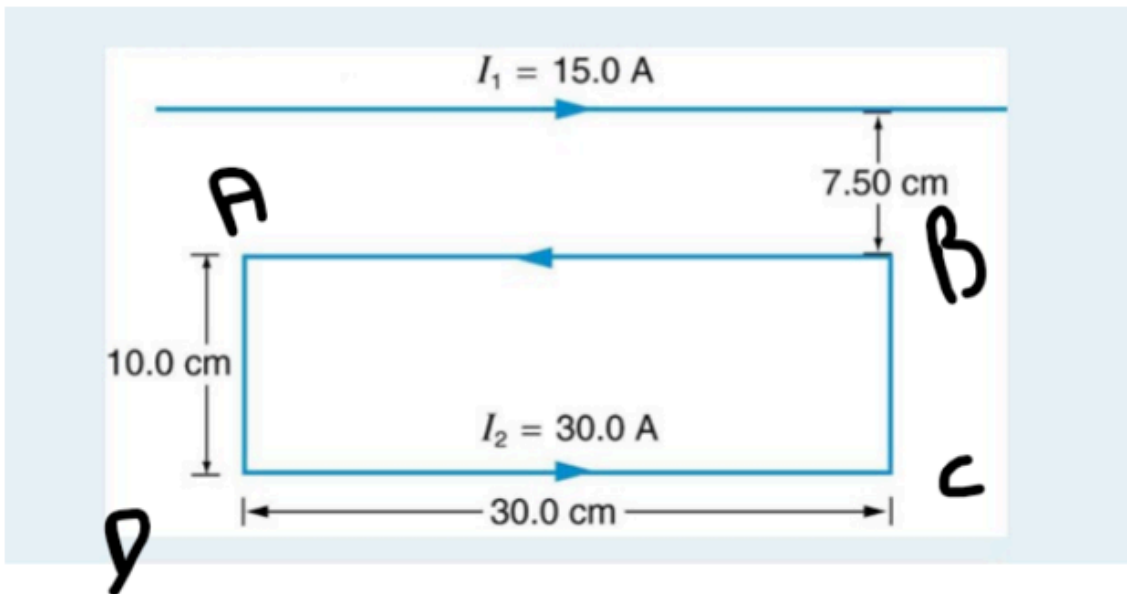
CW



Can not be determined



Question 2 15 pts



Consider this configuration. The goal is to find the net force acting on the rectangular loop of current (by the straight wire)

There is no force acting on AD and BC. (the magnetic field produced by the straight wire is // or anti // to the segment. There is a force only if B perpendicular to a current carrying segment.

What is the net force applied to the loop. Up is + and down is -

☐

$2 \cdot 10^{-4} \text{ N}$

☐

$-5 \cdot 10^{-4} \text{ N}$

☐

$-2 \cdot 10^{-4} \text{ N}$

☐

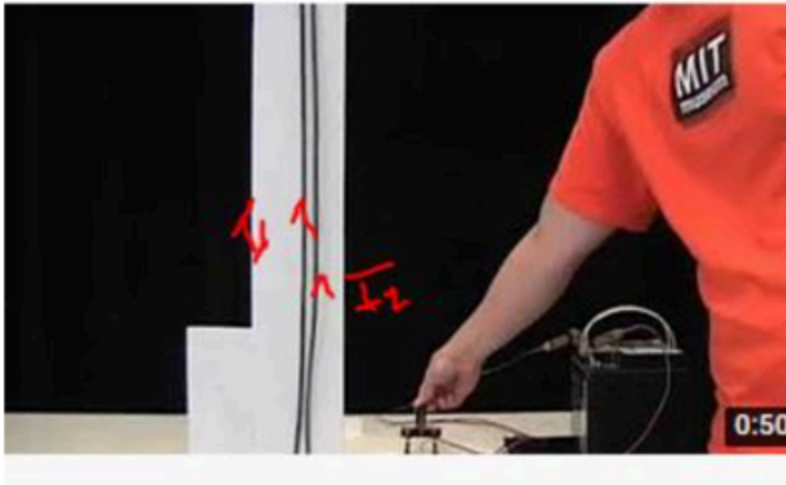
0

☐

$5 \cdot 10^{-4} \text{ N}$

☐

Question 3 15 pts



Those cables are connected to a DC battery 12 V. the current flow in the same direction. I showed the video in class. The cables

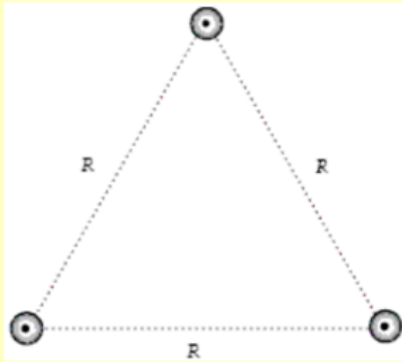
- ☐ there is no force
- ☐ repel each other
- ☐ attract each other
- ☐ can not be determined



Question 4 15 pts

What is the magnetic force acting on one of the wires.

3. The figure shows a cross section of three parallel wires each carrying a current of 5.0 A out of the paper. If the distance $R = 6.0$ mm, what is the magnitude of the magnetic force on a 2.0-m length of any one of the wires?



- a. 2.5 mN
- b. 3.3 mN
- c. 2.2 mN
- d. 2.9 mN
- e. 1.7 mN

Round to the nearest 10th.

hints: Slide 18 (source of magnetism) has the equation for the force between 2 wires. $F/L = (\mu_0/2\pi) I_1 I_2 / R$

you are adding vectors. you have 2 vectors to add at one of the vertex of the equilateral triangle. Its an equilateral triangle so the angle is 60 degrees. A bisector divides a corner in 2 (30 degrees)

hint: you add 2 vectors (same magnitude). these vectors are making a 60 degrees.

there is a tutorial

☐

a

☐

b

☐

d

☐

c

☐

e

☐

Question 5 15 pts

A loop of area 0.250 m^2 is in a uniform 0.0200 T magnetic field. If the flux through the loop is $3.21 \times 10^{-3} \text{ T} \cdot \text{m}^2$, what angle does the normal to the plane of the loop make with the direction of the magnetic field? there is a tutorial-video but easy (hint use definition of change of flux $A B \cos(\theta)$)

☐

10 degrees

☐

50 degrees

☐

40 degrees

☐

0



Question 6 15 pts



A square coil, enclosing an area with sides 2.0 cm long, is wrapped with 2500 turns of wire. A uniform magnetic field perpendicular to its plane is turned on and increases to 0.12 T during an interval of 1.0 s. What average voltage is induced in the coil

hint: there is a tutorial-video. Easy.

☐

1.2V

☐

2.5V

☐

0.12 V

☐

0.22V



Question 7 15 pts

A planar loop consisting of four turns of wire, each of which encloses 250 cm^2 , is oriented perpendicularly to a magnetic field that increases uniformly in magnitude from 10 mT to 25 mT in a time of 5.0 ms. What is the resulting induced current in the coil if the resistance of the coil is 5.0 ohms?

hint: You have the resistance and you find the emf. $I = \text{emf}/R$ (ohms law).

$\text{emf} = NA \text{ change}(B)/\text{change}(\text{time})$

☐

60mA

☐

0.24mA

☐

12mA

☐

48mA

Quiz saved at 6:40pm

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